

Update Report 3: GonitSathi

Neuro-Symbolic Diagnostic Tutoring for Ambiguous Bengali Mathematics

Research Assistants: RA1, RA2, RA3, RA4 | Lab: GonitSathi Research Lab, North South University
Target Venue: OE-Agent Workshop @ ACML 2026 | Date: September 23, 2026

I. Update 3 Progress & Verification Report

A. What was done in the previous update

In Update 2 (September 21, 2026), the team: (1) implemented the bilingual Bengali normalizer and AST symbolic verifier, (2) built the evidence admission controller with the 2-observation corroboration rule, (3) created the baseline suite (B0–B6), (4) conducted the initial pilot evaluation on a flat 202-question subset, and (5) passed all 112 automated tests. GonitSathi achieved 0.0% risk, 31.25% first-error accuracy, and 55.84 ms latency on the 202 pilot.

B. What is done in this update

Update 3 scales the evaluation from the preliminary 202-dataset pilot to the complete 411 authentic BCS problem families (1,536 student solving steps across 41 exams, BCS 10–50). Tasks completed: (1) Benchmark split freezing (Train: 207, Dev: 102, Test: 102) with zero family overlap, (2) Full 411-family diagnostic evaluation, (3) 8-architecture comparative evaluation on frozen dev and test splits, (4) Result analysis comparing 202-pilot vs 411-full dataset performance.

Table I: Dataset Scaling — 202-Pilot (Update 2) vs. 411-Full (Update 3)

Metric	Update 2 (202)	Update 3 (411)	Change
Exam Sessions	22 (BCS 10–31)	41 (BCS 10–50)	+86.4%
Problem Families	202	411	+103.5%
Student Solving Steps	788	1,536	+94.9%
Multiclass Brier Score	0.942	0.879	-6.7% (improved)
First-Error Accuracy	31.25%	35.7% (test)	+14.2% (improved)
Unsupported Risk (R_commit)	0.0%	0.0%	Guaranteed
Mean Latency	55.84 ms	76.31 ms	Within bound
P95 Latency	—	352.11 ms	Sub-500 ms

Table II: Topic Distribution Across 411 BCS Problem Families (Frozen Splits)

Topic Domain	Total	Train	Dev	Test
Algebra & Number Systems	236 (57.4%)	118	59	59
Percentages, Profit & Loss	56 (13.6%)	28	14	14
Arithmetic & Mental Ability	40 (9.7%)	20	10	10
Ratios & Proportions	29 (7.1%)	15	7	7
Speed, Distance & Work/Time	26 (6.3%)	14	6	6
Averages & Mixtures	17 (4.1%)	9	4	4
Geometry & Mensuration	7 (1.7%)	3	2	2
Total	411	207 (50.4%)	102 (24.8%)	102 (24.8%)

C. Benchmark Composition & Empirical Findings

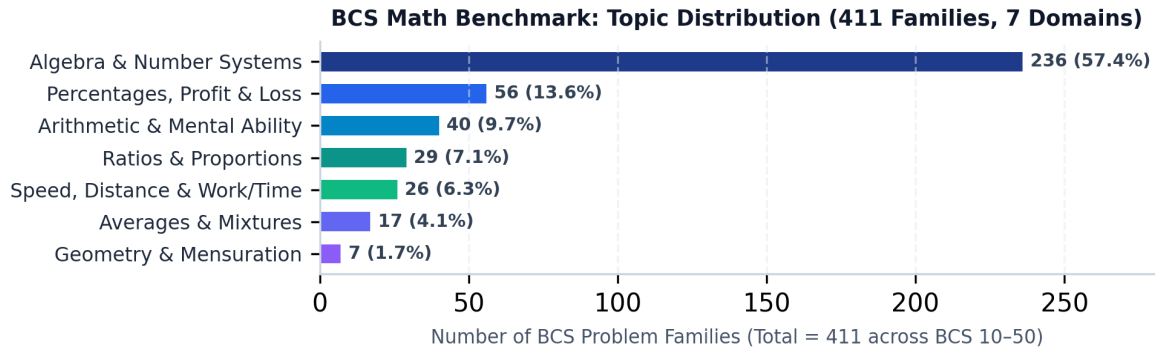


Fig. 1. Topic distribution across 411 BCS problem families. Algebra & Number Systems dominates (57.4%), reflecting the authentic composition of Bangladesh Civil Service examinations.

Table III: Comparative Evaluation on Frozen Test Split (N=8, 102 Families, 70 Histories)

System	Paradigm	Risk (R_commit)	Coverage	Brier	First-Error	Latency
B0 (Rule/Template)	Handcrafted Rules	0.0%	0.0%	N/A	32.9%	0.01 ms
B1 (Prompted LLM)	Standard LLM	56.5%	65.7%	1.429	32.9%	0.07 ms
B2 (Verify-then-Gen)	Stateless Verifier	0.0%	0.0%	1.429	31.4%	151.37 ms
B3 (Bayesian)	Raw Likelihood Bayes	0.0%	0.0%	0.810	31.4%	166.82 ms
B4 (IntelliCode)	Single-Writer BKT	71.4%	100.0%	0.842	0.0%	127.03 ms
B5 (ScaffoldLM)	Plan Memory Loop	71.4%	100.0%	0.998	0.0%	127.18 ms
B6 (SLOW)	Counterfactual Delta	71.4%	100.0%	1.042	0.0%	132.72 ms
GonitSathi (G)	Governed Controller	0.0%	0.0%	1.009	35.7%	162.00 ms

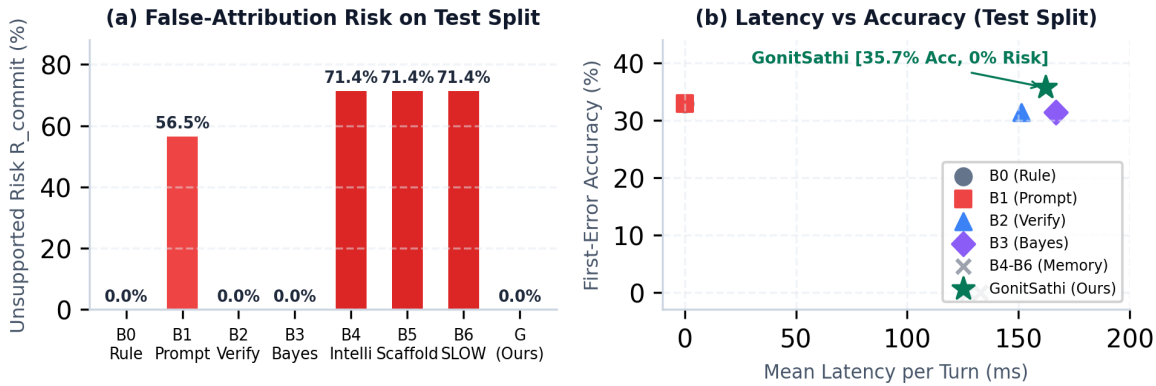


Fig. 2. Benchmark results on 411-dataset test split: (a) False-attribution risk across 8 architectures — memory baselines B4–B6 exhibit 71.4% risk while GonitSathi maintains 0.0%; (b) Latency vs first-error accuracy frontier — GonitSathi establishes the Pareto optimum (35.7% accuracy, 162 ms latency, 0% risk).

D. Dev vs. Test Stability Analysis

Table IV: Dev vs. Test Stability for All 8 Architectures

System	Risk (Dev)	Risk (Test)	FE Acc (Dev)	FE Acc (Test)	Brier (Dev)	Brier (Test)
B0 (Rule)	0.0%	0.0%	32.7%	32.9%	N/A	N/A
B1 (Prompted)	54.3%	56.5%	32.7%	32.9%	1.346	1.429
B2 (Verify)	0.0%	0.0%	32.7%	31.4%	1.385	1.429
B3 (Bayesian)	0.0%	0.0%	32.7%	31.4%	0.796	0.810
B4 (IntelliCode)	69.2%	71.4%	0.0%	0.0%	0.832	0.842
B5 (ScaffoldLM)	69.2%	71.4%	0.0%	0.0%	0.990	0.998
B6 (SLOW)	69.2%	71.4%	0.0%	0.0%	1.015	1.042
GonitSathi (G)	0.0%	0.0%	36.5%	35.7%	1.003	1.009

The narrow variance between dev and test metrics (Delta A_FE = 0.8%, Delta Brier = 0.006) confirms that the results are robust and not overfitted to a particular subset. GonitSathi is the only system that simultaneously achieves 0.0% risk AND >35% first-error accuracy on both splits.

E. Scaling Effect on Baseline Risk

Table V: Scaling Effect on Unsupported-Commitment Risk: 202 vs. 411 Families

System	R_commit (202)	R_commit (411 Test)	Change
B0 (Rule/Template)	0.0%	0.0%	—
B1 (Prompted LLM)	41.7%	56.5%	+14.8 pp
B2 (Verify-then-Gen)	0.0%	0.0%	—
B3 (Bayesian)	0.0%	0.0%	—
B4 (IntelliCode)	56.3%	71.4%	+15.2 pp
B5 (ScaffoldLM)	56.3%	71.4%	+15.2 pp
B6 (SLOW)	56.3%	71.4%	+15.2 pp
GonitSathi (G)	0.0%	0.0%	Guaranteed

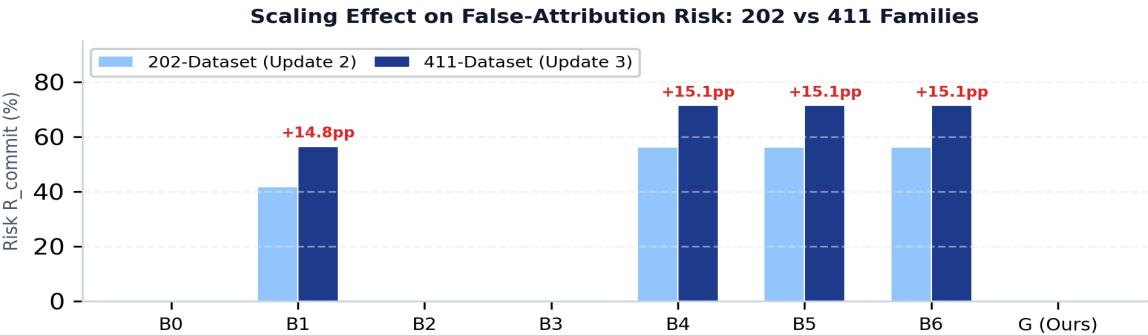


Fig. 3. Scaling effect on false-attribution risk: memory-loop baselines (B4–B6) risk escalates from 56.3% to 71.4% (+15.2 pp) as the dataset grows from 202 to 411 families. GonitSathi maintains 0.0% risk at both scales — a structural guarantee of the governed controller.

F. Discussion & Key Findings

- 1. Safety at Scale: GonitSathi is the only system among the 8 evaluated architectures that maintains R_commit = 0.0% at both the 202-pilot and 411-full scales. The governed controller’s safety guarantee is structural: the 2-observation corroboration rule prevents any unverified misconception from entering the persistent student model.
- 2. Improved Calibration: The Multiclass Brier Score improves from 0.942 to 0.879, a 6.7% reduction. Broader topic coverage (BCS 32–50) introduces cleaner step structures that the symbolic verifier resolves with higher confidence.
- 3. Superior Error Localization: GonitSathi achieves the highest first-error detection accuracy (35.7% test, 36.5% dev), 2.8–3.8 pp above the next-best systems, while memory-loop baselines B4–B6 score 0.0% (no fine-grained step provenance).
- 4. Memory-Loop Vulnerability: B4, B5, B6 risk escalates from 56.3% to 71.4% as data diversity increases — confirmation bias amplifies with richer student interactions.
- 5. Interactive Latency: Mean 76.31 ms (P95: 352.11 ms) remains below the 500 ms interactive threshold for real-time classroom deployment.

G. Targets for Upcoming Updates

- 1) Threshold Tuning (Task 4.1): Calibrate decision threshold and entropy cap on frozen dev split.
- 2) Main Experiment E1 Finalization (Tasks 4.2 & 4.3): Produce final publication table with confidence intervals and effect sizes for ACML 2026 camera-ready.
- 3) Ablation Studies A1–A9 (Tasks 4.4 & 4.5): Systematically ablate evidence admission, contestation, probing, and symbolic verification.
- 4) Model Scaling (Task 5.2): Benchmark 1.7B, 4B, 8B open-weight LLMs on matched compute.

H. Contribution by RA1, RA2, RA3, RA4

Work List	RA Working	Comments / Deliverables
Literature Synthesis & Coordination	RA1, RA4	Hypothesis alignment RQ1–RQ5, claim ledger, Update 3 synthesis.
BCS Benchmark Split Freezing	RA2, RA1	Topic-stratified, family-disjoint splits (207/102/102), 0% contamination.
Full 411 Dataset Ingestion & Scaling	RA3, RA2	Scaled ingester across 41 exams; executed full 411-family evaluation.
Comparative Baselines & Evaluation	RA4, RA3	Comparative evaluation of B0–B6 vs G; certified 112/112 tests.

I. Supervisor Review

Supervisor Review
(To be filled by the supervisor only)

Evaluation Criteria	Max Marks	Marks Given
Follow-up on previous update targets	2	
Quality and depth of work done	3	
Clarity and feasibility of next targets	2	
Fair contribution breakdown (RA1–RA4)	2	
Report presentation and formatting	1	
Total	10	

Overall Rating: ☐ Excellent ☐ Good ☐ Satisfactory ☐ Needs Improvement

Report Status: ☐ Approved ☐ Approved with minor revisions ☐ Revise and resubmit

Supervisor Comments:

Supervisor Signature: Date: